# Class 18: Mini Project

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### Background

Pertussis (a.k.a. whooping cough) is a common lung infection caused by the bacteria B. Pertussis.

This can affect people of all ages but it is most sever for those under 1 year of age.

Due to its seriousness the CDC track the number of reported cases in the US: CDC We can "scrape" this data with **datapasta** package.

```
cdc <- data.frame(
  year = c(
    1922L,
    1923L,
    1924L,
    1925L,
    1926L,
    1927L,
    1928L,
    1929L,
    1930L,
    1931L,
    1932L,
    1933L,
    1934L,
```

```
    1935L,
    1936L,
    1937L,
    1938L,
    1939L,
    1940L,
    1941L,
    1942L,
    1943L,
    1944L,
    1945L,
    1946L,
    1947L,
    1948L,
    1949L,
    1950L,
    1951L,
    1952L,
    1953L,
    1954L,
    1955L,
    1956L,
    1957L,
    1958L,
    1959L,
    1960L,
    1961L,
    1962L,
    1963L,
    1964L,
    1965L,
    1966L,
    1967L,
    1968L,
    1969L,
    1970L,
    1971L,
    1972L,
    1973L,
    1974L,
    1975L,
    1976L,
```

```
    1977L,
    1978L,
    1979L,
    1980L,
    1981L,
    1982L,
    1983L,
    1984L,
    1985L,
    1986L,
    1987L,
    1988L,
    1989L,
    1990L,
    1991L,
    1992L,
    1993L,
    1994L,
    1995L,
    1996L,
    1997L,
    1998L,
    1999L,
    2000L,
    2001L,
    2002L,
    2003L,
    2004L,
    2005L,
    2006L,
    2007L,
    2008L,
    2009L,
    2010L,
    2011L,
    2012L,
    2013L,
    2014L,
    2015L,
    2016L,
    2017L,
    2018L,
```

```
    2019L,
    2020L,
    2021L,
    2022L,
    2023L,
    2024L,
    2025L
  ),
  cases = c(
    107473,
    164191,
    165418,
    152003,
    202210,
    181411,
    161799,
    197371,
    166914,
    172559,
    215343,
    179135,
    265269,
    180518,
    147237,
    214652,
    227319,
    103188,
    183866,
    222202,
    191383,
    191890,
    109873,
    133792,
    109860,
    156517,
    74715,
    69479,
    120718,
    68687,
    45030,
    37129,
    60886,
```

```
    62786,
    31732,
    28295,
    32148,
    40005,
    14809,
    11468,
    17749,
    17135,
    13005,
    6799,
    7717,
    9718,
    4810,
    3285,
    4249,
    3036,
    3287,
    1759,
    2402,
    1738,
    1010,
    2177,
    2063,
    1623,
    1730,
    1248,
    1895,
    2463,
    2276,
    3589,
    4195,
    2823,
    3450,
    4157,
    4570,
    2719,
    4083,
    6586,
    4617,
    5137,
    7796,
```

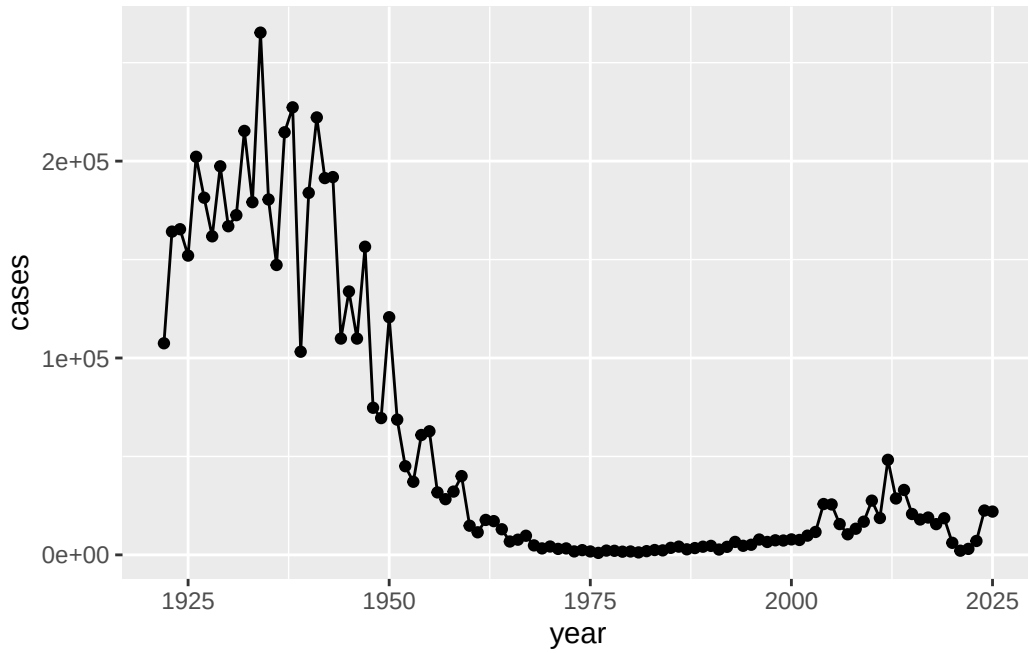
```
    6564,
    7405,
    7298,
    7867,
    7580,
    9771,
    11647,
    25827,
    25616,
    15632,
    10454,
    13278,
    16858,
    27550,
    18719,
    48277,
    28639,
    32971,
    20762,
    17972,
    18975,
    15609,
    18617,
    6124,
    2116,
    3044,
    7063,
    22538,
    21996
  )
)
```

Q.1 Make a plot of `year` vs `cases`

```
library(ggplot2)
```

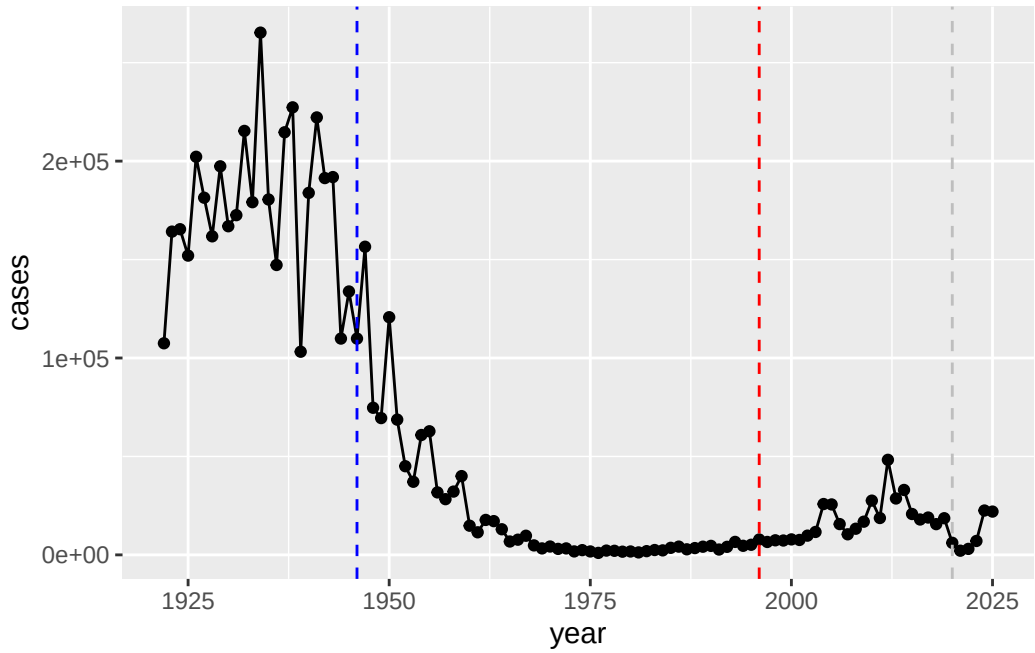
```
Warning: package 'ggplot2' was built under R version 4.5.2
```

```
ggplot(cdc) +
  aes(year, cases) +
  geom_point() +
  geom_line()
```




Q2. Add some major milestones including the first wP vaccine roll-out (1946), the switch to the newer aP vaccine (1996). The covid years (2020).

```
ggplot(cdc) +
  aes(year, cases) +
  geom_point() +
  geom_line() +
  geom_vline(xintercept = 1946, col="blue", lty=2) +
  geom_vline(xintercept = 1996, col="red", lty=2) +
  geom_vline(xintercept = 2020, col="grey", lty=2)
```




There were high case numbers in the pre 1940s because there was no widespread pertussis vaccination yet, so the population had little immunity and the bacteria spread easily between people.

> Q3. aP induced protection wanes faster than wP, why?

aP vaccine protection wanes faster because it contains only a few purified pertussis antigens, producing a narrower and weaker immune response compared to the broader immune stimulation caused by the wP vaccine.

**Why is this vaccine-preventable disease on the upswing?** To answer this question we need to investigate the mechanisms underlying waning protection against pertussis. This requires evaluation of pertussis-specific immune responses over time in wP and aP vaccinated individuals.

The new and ongoing CMI-PB project aims to provide the scientific community with this very information.

## CMI-PB project

Computational Models of Immunity - Pertussis Boost aims to provide the scientific community with this very information.

They make there data avalible via JSON format APIs. We can read this in R with the `read_json()` function from the **jsonlite** package:

```r
library(jsonlite)

subject <- read_json("https://www.cmi-pb.org/api/v5_1/subject",
                     simplifyVector = TRUE)

head(subject)
```

```
  subject_id infancy_vac biological_sex              ethnicity  race
1          1          wP         Female Not Hispanic or Latino White
2          2          wP         Female Not Hispanic or Latino White
3          3          wP         Female                Unknown White
4          4          wP           Male Not Hispanic or Latino Asian
5          5          wP           Male Not Hispanic or Latino Asian
6          6          wP         Female Not Hispanic or Latino White
  year_of_birth date_of_boost      dataset
1    1986-01-01    2016-09-12 2020_dataset
2    1968-01-01    2019-01-28 2020_dataset
3    1983-01-01    2016-10-10 2020_dataset
4    1988-01-01    2016-08-29 2020_dataset
5    1991-01-01    2016-08-29 2020_dataset
6    1988-01-01    2016-10-10 2020_dataset
```

Q4. How many "wP" and "aP" individuals are in this `subject` table?

```r
table(subject$infancy_vac)
```

```
aP wP
87 85
```

Q5. What is the biological sex breakdown?

```r
table(subject$biological_sex)
```

```
Female   Male
   112     60
```

Q6. In terms of race and gender is this dataset representative of the US population?

```r
table(subject$race, subject$biological_sex)
```

```
                                          Female Male
  American Indian/Alaska Native                0    1
  Asian                                       32   12
  Black or African American                    2    3
  More Than One Race                          15    4
  Native Hawaiian or Other Pacific Islander    1    1
  Unknown or Not Reported                     14    7
  White                                       48   32
```

Let's read some more database tables:

```r
specimen <- read_json("https://www.cmi-pb.org/api/v5_1/specimen",
                      simplifyVector = TRUE)
ab_titer <- read_json("https://www.cmi-pb.org/api/v5_1/plasma_ab_titer",
                      simplifyVector = TRUE)
```

```r
head(specimen)
```

```
  specimen_id subject_id actual_day_relative_to_boost
1           1          1                           -3
2           2          1                            1
3           3          1                            3
4           4          1                            7
5           5          1                           11
6           6          1                           32
  planned_day_relative_to_boost specimen_type visit
1                             0         Blood     1
2                             1         Blood     2
3                             3         Blood     3
4                             7         Blood     4
5                            14         Blood     5
6                            30         Blood     6
```

```r
head(ab_titer)
```

```
  specimen_id isotype is_antigen_specific antigen        MFI MFI_normalised
```

```
1           1     IgE               FALSE   Total 1110.21154       2.493425
2           1     IgE               FALSE   Total 2708.91616       2.493425
3           1     IgG                TRUE      PT   68.56614       3.736992
4           1     IgG                TRUE     PRN  332.12718       2.602350
5           1     IgG                TRUE     FHA 1887.12263      34.050956
6           1     IgE                TRUE     ACT    0.10000       1.000000
   unit lower_limit_of_detection
1 UG/ML                 2.096133
2 IU/ML                29.170000
3 IU/ML                 0.530000
4 IU/ML                 6.205949
5 IU/ML                 4.679535
6 IU/ML                 2.816431
```

To analyze this data we need to first "join" the different tables so we have all the data in one place and not spread across different tables.

We can use the `*join()` family of functions from **dplyr** to do this.

```r
library(dplyr)
```

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```r
meta <- inner_join(subject, specimen)
```

```
Joining with `by = join_by(subject_id)`
```

```r
head(meta)
```

```
  subject_id infancy_vac biological_sex              ethnicity  race
1          1          wP         Female Not Hispanic or Latino White
2          1          wP         Female Not Hispanic or Latino White
3          1          wP         Female Not Hispanic or Latino White
4          1          wP         Female Not Hispanic or Latino White
5          1          wP         Female Not Hispanic or Latino White
6          1          wP         Female Not Hispanic or Latino White
  year_of_birth date_of_boost      dataset specimen_id
1    1986-01-01    2016-09-12 2020_dataset           1
2    1986-01-01    2016-09-12 2020_dataset           2
3    1986-01-01    2016-09-12 2020_dataset           3
4    1986-01-01    2016-09-12 2020_dataset           4
5    1986-01-01    2016-09-12 2020_dataset           5
6    1986-01-01    2016-09-12 2020_dataset           6
  actual_day_relative_to_boost planned_day_relative_to_boost specimen_type
1                           -3                             0         Blood
2                            1                             1         Blood
3                            3                             3         Blood
4                            7                             7         Blood
5                           11                            14         Blood
6                           32                            30         Blood
  visit
1     1
2     2
3     3
4     4
5     5
6     6
```

```r
abdata <- inner_join(ab_titer, meta)
```

```
Joining with `by = join_by(specimen_id)`
```

```r
head(abdata)
```

```
  specimen_id isotype is_antigen_specific antigen        MFI MFI_normalised
1           1     IgE               FALSE   Total 1110.21154       2.493425
2           1     IgE               FALSE   Total 2708.91616       2.493425
3           1     IgG                TRUE      PT   68.56614       3.736992
4           1     IgG                TRUE     PRN  332.12718       2.602350
5           1     IgG                TRUE     FHA 1887.12263      34.050956
```

```
6           1     IgE                TRUE     ACT    0.10000       1.000000
   unit lower_limit_of_detection subject_id infancy_vac biological_sex
1 UG/ML                 2.096133          1          wP         Female
2 IU/ML                29.170000          1          wP         Female
3 IU/ML                 0.530000          1          wP         Female
4 IU/ML                 6.205949          1          wP         Female
5 IU/ML                 4.679535          1          wP         Female
6 IU/ML                 2.816431          1          wP         Female
               ethnicity  race year_of_birth date_of_boost      dataset
1 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
2 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
3 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
4 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
5 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
6 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
  actual_day_relative_to_boost planned_day_relative_to_boost specimen_type
1                           -3                             0         Blood
2                           -3                             0         Blood
3                           -3                             0         Blood
4                           -3                             0         Blood
5                           -3                             0         Blood
6                           -3                             0         Blood
  visit
1     1
2     1
3     1
4     1
5     1
6     1
```

Q7. What Antibody isotypes are measured for these patients?

```
table(abdata$isotype)
```

```
  IgE   IgG  IgG1  IgG2  IgG3  IgG4
 6698  7265 11993 12000 12000 12000
```

Q8. What antigens are reported?

```
table(abdata$antigen)
```

```
    ACT   BETV1      DT   FELD1     FHA  FIM2/3   LOLP1     LOS Measles     OVA
   1970    1970    6318    1970    6712    6318    1970    1970    1970    6318
    PD1     PRN      PT     PTM   Total      TT
   1970    6712    6712    1970     788    6318
```

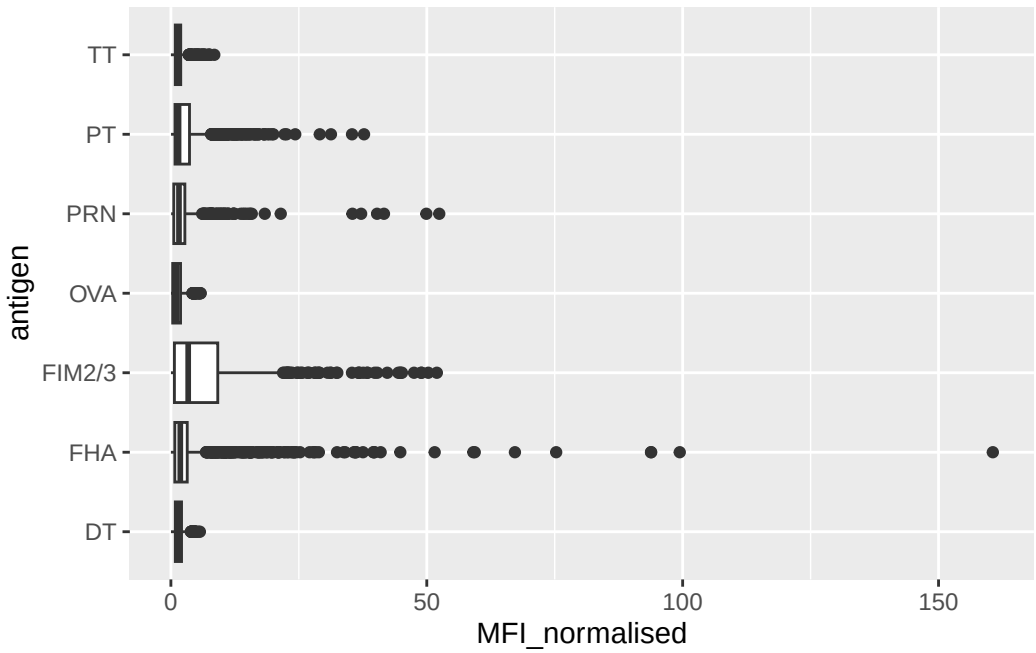
Let's focus on the IgG antigens and make a plot of MFI_normalized for all antigens.

```r
igg <- abdata |>
  filter(isotype == "IgG")
head(igg)
```

```
  specimen_id isotype is_antigen_specific antigen        MFI MFI_normalised
1           1     IgG                TRUE      PT   68.56614       3.736992
2           1     IgG                TRUE     PRN  332.12718       2.602350
3           1     IgG                TRUE     FHA 1887.12263      34.050956
4          19     IgG                TRUE      PT   20.11607       1.096366
5          19     IgG                TRUE     PRN  976.67419       7.652635
6          19     IgG                TRUE     FHA   60.76626       1.096457
   unit lower_limit_of_detection subject_id infancy_vac biological_sex
1 IU/ML                 0.530000          1          wP         Female
2 IU/ML                 6.205949          1          wP         Female
3 IU/ML                 4.679535          1          wP         Female
4 IU/ML                 0.530000          3          wP         Female
5 IU/ML                 6.205949          3          wP         Female
6 IU/ML                 4.679535          3          wP         Female
               ethnicity  race year_of_birth date_of_boost      dataset
1 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
2 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
3 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
4                Unknown White    1983-01-01    2016-10-10 2020_dataset
5                Unknown White    1983-01-01    2016-10-10 2020_dataset
6                Unknown White    1983-01-01    2016-10-10 2020_dataset
  actual_day_relative_to_boost planned_day_relative_to_boost specimen_type
1                           -3                             0         Blood
2                           -3                             0         Blood
3                           -3                             0         Blood
4                           -3                             0         Blood
5                           -3                             0         Blood
6                           -3                             0         Blood
  visit
1     1
```

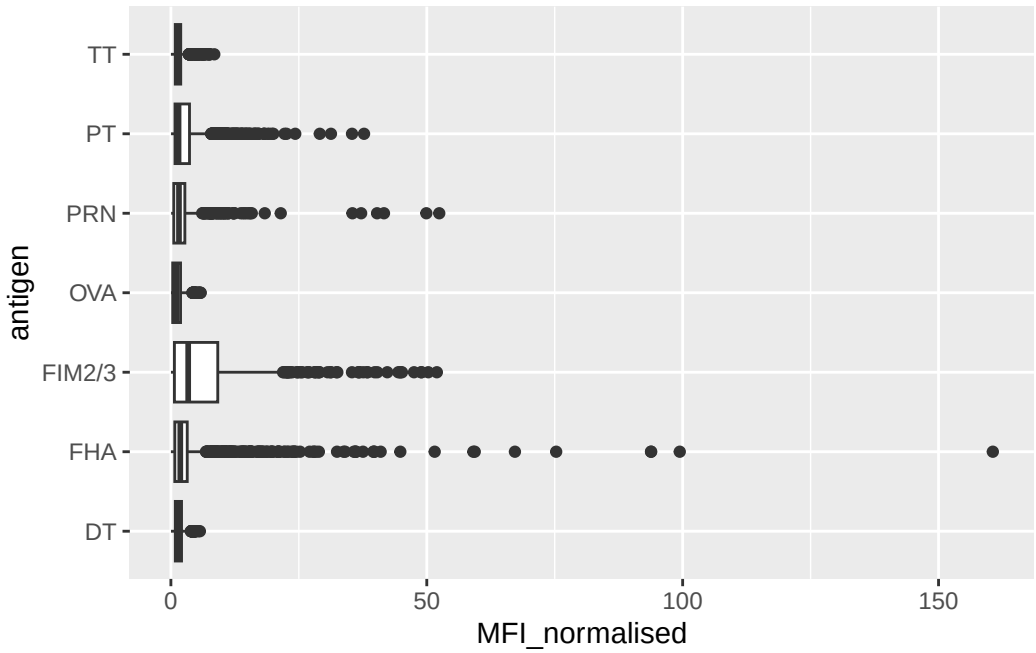
```
2      1
3      1
4      1
5      1
6      1
```

```
ggplot(igg) +
  aes(MFI_normalised, antigen) +
  geom_boxplot()
```




Q9. Is there a difference for aP vs wP individuals with these values?

```
ggplot(igg) +
  aes(MFI_normalised, antigen) +
  geom_boxplot()
```

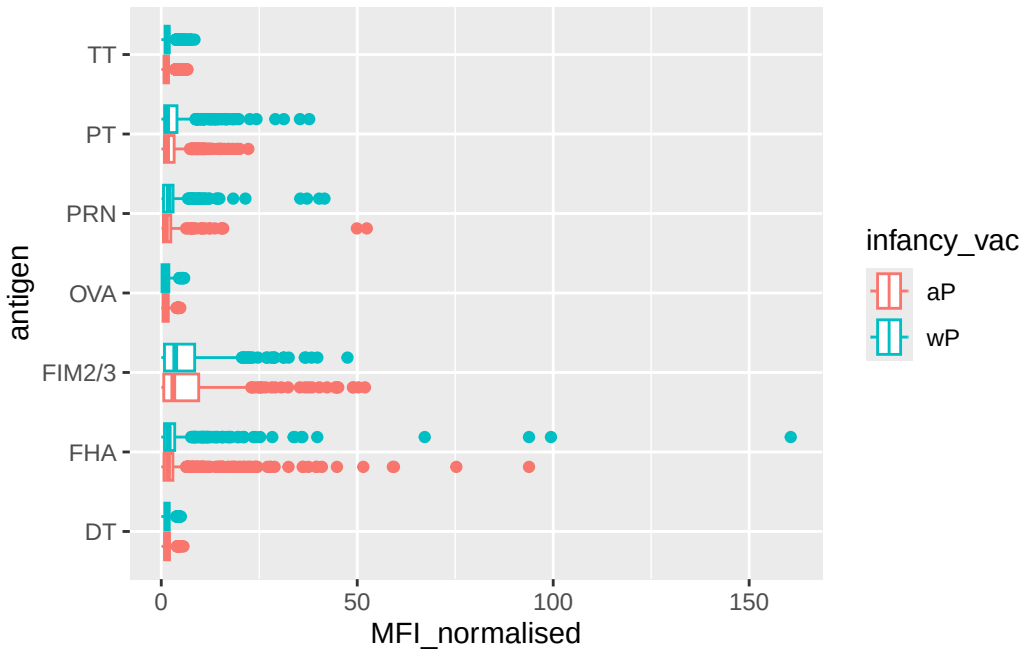



```
  facet_wrap(~infancy_vac)
```

```
<ggproto object: Class FacetWrap, Facet, gg>
    attach_axes: function
    attach_strips: function
    compute_layout: function
    draw_back: function
    draw_front: function
    draw_labels: function
    draw_panel_content: function
    draw_panels: function
    finish_data: function
    format_strip_labels: function
    init_gtable: function
    init_scales: function
    map_data: function
    params: list
    set_panel_size: function
    setup_data: function
    setup_panel_params: function
    setup_params: function
    shrink: TRUE
```

```
    train_scales: function
    vars: function
    super:  <ggproto object: Class FacetWrap, Facet, gg>
```

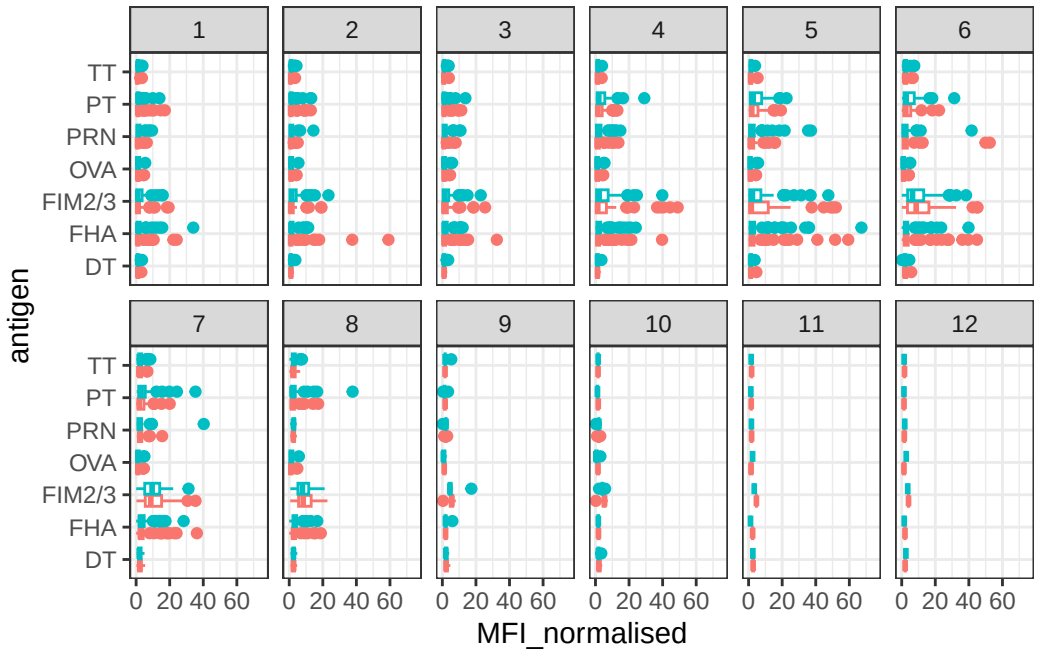
```
ggplot(igg) +
  aes(MFI_normalised, antigen, col=infancy_vac) +
  geom_boxplot()
```




Q10. Is there a temprol response - i.e do values increase or decrease over time?

```
ggplot(igg) +
  aes(MFI_normalised, antigen, col=infancy_vac ) +
  geom_boxplot(show.legend = FALSE) +
  facet_wrap(vars(visit), nrow=2) +
  xlim(0,75) +
  theme_bw()
```

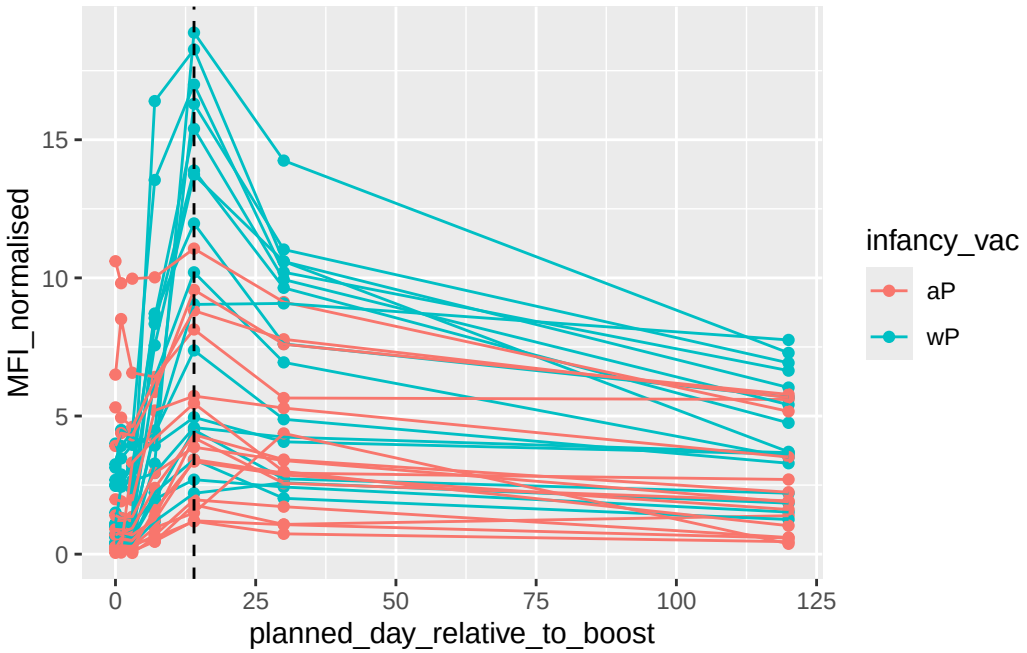
```
Warning: Removed 5 rows containing non-finite outside the scale range
(`stat_boxplot()`).
```




### Focus on "PT" Pertusisis Toxin antigen

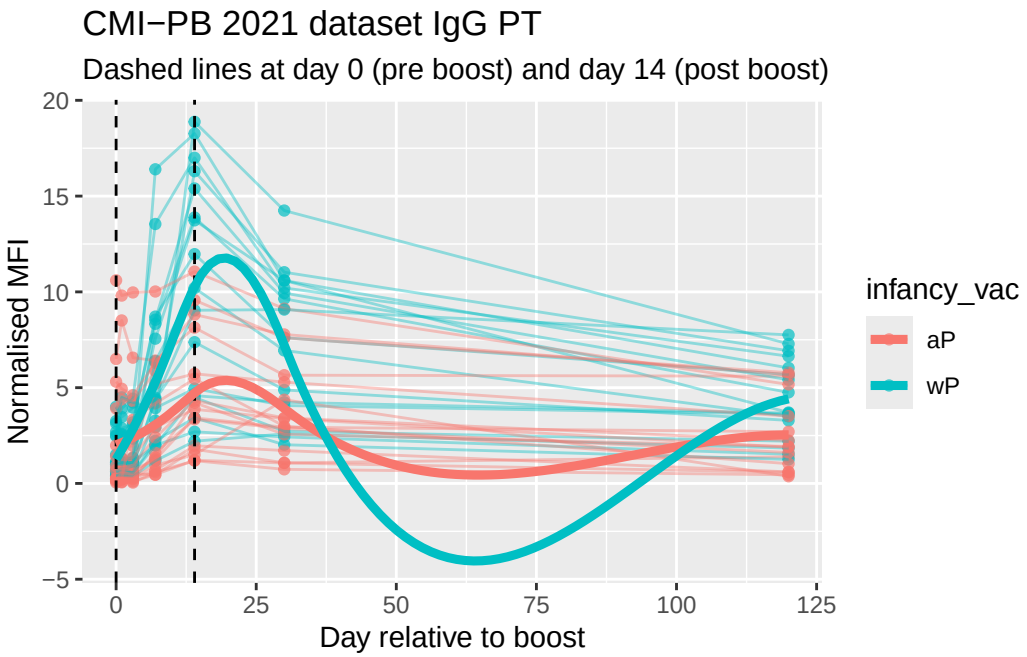
```
pt.igg.21 <- igg |> filter(antigen =="PT",
              dataset == "2021_dataset" )
```

```
ggplot(pt.igg.21) +
  aes(planned_day_relative_to_boost,
      MFI_normalised,
      col=infancy_vac,
      group = subject_id) +
  geom_point() +
  geom_line() +
  geom_vline(xintercept = 14, lty=2)
```




```
ggplot(pt.igg.21,
       aes(x = planned_day_relative_to_boost,
           y = MFI_normalised,
           color = infancy_vac,
           group = subject_id)) +
  geom_point(alpha = 0.7) +
  geom_line(alpha = 0.4) +
  geom_vline(xintercept = c(0, 14), linetype = 2) +
  geom_smooth(aes(group = infancy_vac),
              se = FALSE,
              linewidth = 1.6) +
  labs(title = "CMI-PB 2021 dataset IgG PT",
       subtitle = "Dashed lines at day 0 (pre boost) and day 14 (post boost)",
       x = "Day relative to boost",
       y = "Normalised MFI")
```

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```


CMI−PB 2021 dataset IgG PT
Dashed lines at day 0 (pre boost) and day 14 (post boost)
Normalised MFI
20
15
10
5
0
−5
0
25
50
75
100
125
Day relative to boost
infancy_vac
aP
wP